HUSAM AHMED

Ottawa, ON |husamdaifalla01@gmail.com |(647) 685-6908 |linkedin.com/in/husam-ahmed2025

May 14, 2026

Hiring Manager
Health Technology and Systems Team
Canada's Drug Agency (CDA-AMC)

Re: Clinical Research Assistant (Temporary)

Dear Hiring Manager,

I'm applying for the Clinical Research Assistant role because the day-to-day work, screening literature, extracting data from studies, drafting reports, contributing to committee outputs, parallels what I'm already doing at Stelis Environmental Solutions assessing a new microbial monitoring technology, just applied to a different domain.

At Stelis Environmental Solutions, I'm the junior researcher on the team's assessment of ColiMinder, a relatively new real-time enzymatic microbial monitoring technology, as a possible supplement or replacement for culture-based drinking-water testing. My work feeds analytical findings and recommendations to project leads on three recurring business issues that map onto how a health technology assessment is structured: market prioritization, technology validation, and regulatory exposure.

For market prioritization, I built and maintain Power BI heat maps of microbial, chemical, lead, and nitrate exceedances across Canadian and US jurisdictions from public Health Canada, US EPA, and provincial reports. The analysis goes to project leads as a recommendation on which jurisdictions match the team's solution profile and which to deprioritize, feeding the go-to-market direction.

For technology validation, I compare ColiMinder real-time enzymatic outputs against culture-based lab references in Excel, calculate percent agreement across pilot sites, and advise project leads on validation conclusions and threshold or sampling-frequency adjustments. My pivot-table analysis of alarm frequency and false-positive rates surfaces seasonal and site-level patterns the team uses to tune the technology.

For regulatory exposure, I scan Health Canada GCDWQ, Ontario Regulations 169/03 and 170/03, and US EPA drinking-water rules alongside peer-reviewed literature on enzymatic β-glucuronidase detection to identify where the technology is accepted and the gating criteria. The consolidated SOP and regulatory change log I maintain keep procedures current as federal and provincial rules evolve.

At IRCC's Operations Support Centre, my parallel role is investigative case processing under the Immigration and Refugee Protection Act (IRPA). When a biometric failure arrives from one of 60+ overseas Visa Application Centres or Missions, I read the trace, interpret partner-system responses, and weigh the resolution path against IRPA's biometric and information-handling provisions before documenting the reasoning so each file holds full context.

My BSc in Biology from the University of Ottawa is the foundation: microbiology coursework, statistical analysis of lab datasets in Excel, and technical lab reports laying out methods, results, and limitations. I haven't done formal systematic review work yet, and the chance to learn that under senior team members at CDA-AMC is one of the things that drew me to this role.

Outside of work, I built curious-brain to find out whether neuroscience principles can actually govern how a software system reasons, not just name its parts. It's a personal continuation of an interest from my BSc Biology coursework, with each layer (memory consolidation, predictive coding, attention gating) implemented as a design constraint drawn from primary literature: Friston on the free energy principle, Lisman and Jensen on theta-gamma coupling, Buzsaki on hippocampal replay during sleep. It's been my way of actually understanding the papers rather than just reading them.

I'm bilingual in English and Arabic with working knowledge of French, eligible for federal security clearance, and currently managing two simultaneous roles. I'd value the chance to bring this combination to the Health Technology and Systems team.

Thank you for your consideration.

Sincerely,

Husam Ahmed